

5.4 Detailed specific example: India



India is a country in South Asia. It has the largest population in the world at 1.43 billion people (around 17% of the world's population). Due to this and its rapid economic development, India is the third largest emitter of greenhouse gases in the world. In 2021, India contributed 7% of global greenhouse gas emissions. However, its large population means that its per capita (per person) emissions are low, at just over 2 tons of carbon every year.

India has committed itself to achieving the United Nation's Sustainable Development Goals by 2030, which includes addressing climate change. However, the country is only likely to achieve two of the seventeen goals. Climate action has stagnated in India.

Task 1

- Draw a line graph** from the data in the table to show India's carbon dioxide emissions per capita, and the average temperature increase between 2000 and 2020. Add a title, label the axes, and draw a key.
- Describe the graph.**
- Suggest reasons for the increase in carbon dioxide emissions per person.

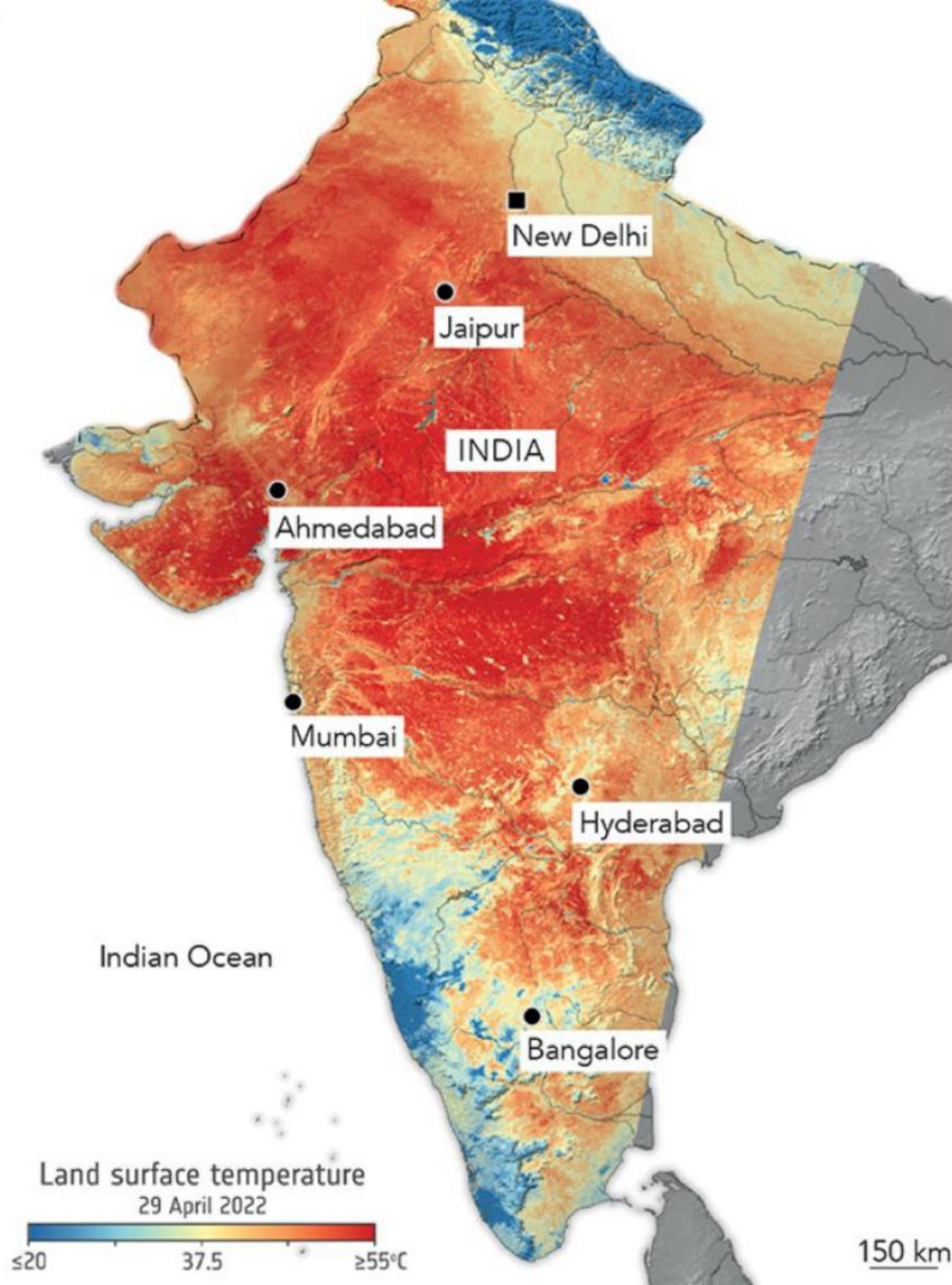
Year	Average temperature (°C)	Per capita CO ₂ emissions (tons)
2000	24.81	0.9
2005	24.81	1.0
2010	25.52	1.4
2015	25.01	1.7
2020	24.90	1.7

TIP

When drawing a graph, think about the scale of the axes so that you present the data clearly and ensure it is not distorted.

Impacts of climate change on India

India is feeling the effects of climate change already. The country has experienced a rise in average temperatures, leading to more frequent and intense heatwaves. This has impacted agriculture, water resources, and public health. In 2022, for example, a heatwave across India caused temperatures to reach over 40 °C. Heatwaves will affect more than 300 million people in India by 2050 and lower the quality of life for almost 600 million Indians by 2100.



Source **A**: A map showing the Indian heatwave in April 2022

TASK 2: Study Source **A**

- Describe the heatwave in India** in April 2022.
- Explain the short-term and long-term impacts that could be linked to this heatwave.

Changing monsoon patterns due to climate change are resulting in variations in rainfall distribution and intensity. This can affect agriculture, as farmers rely heavily on the monsoon for water. In 2023, heavy monsoon rains damaged basmati rice and tomato crops resulting in food inflation. Droughts, too, will impact crop productivity and put additional strain on India's water resources. This could result in food security issues. India is prone to extreme weather events such as cyclones, floods, and droughts. Climate change may increase the frequency and intensity of these events. India is projected to experience a 20% increase in extreme rainfall events.

Rising sea levels pose a threat to low-lying coastal regions. India could lose 1500 square km of land by 2050 due to rising sea levels. This reduces land available for agriculture and building homes. It also means many people become displaced. Access to clean water and sanitation facilities may be compromised during and after flooding, contributing to public health challenges through the spread of diseases. Climate change can also disrupt ecosystems, affecting plant and animal species, resulting in species extinction.

The Himalayan region, which is a crucial source of freshwater for many rivers in India, has witnessed glacial retreat. At least one-third of the glaciers in the region will be gone by 2100. This can affect agriculture and the water supplies of urban areas downstream. Catastrophic floods can also occur if glacial lake dams (water from a melting glacier detained by a dam of deposited sediment) burst. In the state of Sikkim, a glacial lake burst in October 2023, killing over 100 people.



Figure 5.14: A glacial lake dam in Pithahara, Madhya Pradesh, India

While not directly caused by climate change, India also faces major challenges related to air pollution. Air quality can have severe health implications for the population. The majority (51%) of India's electricity generation is produced by burning coal. Burning coal can cause a range of health problems including higher rates of childhood asthma, heart and lung disease, and some cancers.

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Figure 5.15: Tuticorin Thermal Power Station, Tamil Nadu. Burning coal is India's main source of electricity production.



Figure 5.16: India is investing in renewables like the Tehri Dam in Uttarakhand, a hydroelectric dam, to reduce its carbon footprint.

Managing climate change in India

India faces the dual challenge of tackling climate change whilst also addressing rural poverty. Poverty in India is decreasing, but 1 in 10 Indians still live below the international poverty line of \$2.15 per day. India has an extremely youthful population, with 40% of the population being under 25. The actions taken in response to climate change will therefore impact young people in India the most.

India is committed to the Paris Agreement, which aims to keep global warming below 2 °C. This commitment was shown at the COP26 conference in 2021. India is therefore a crucial leader for other lower-middle-income countries. India’s emissions targets are to reduce its emissions-intensity (greenhouse gas emissions produced per unit of a particular activity) by 45% below 2005 levels by 2030, and to increase the share of non-fossil power capacity to 50% by 2030. However, despite installing significant renewable energy sources, India’s dependence on fossil fuels is increasing as coal and gas-fired power plants operate at peak capacity to meet the rise in seasonal electricity demand from record-hot summers. These targets are therefore unsustainable at present.

Techniques

India is actively addressing the impacts of climate change through various initiatives (see Table 5.2). One key example is the International Solar Alliance, in which India collaborates with other countries to promote solar energy. India has also launched ambitious projects like the ‘One Sun One World One Grid’, aiming to harness solar power globally. The vision behind this project is the mantra that ‘the sun never sets’. This initiative aims to connect different regional grids through a common grid that will be used to transfer renewable energy power.

To tackle the urban challenges that result from climate change in India, cities like Mumbai are investing in sustainable infrastructure, such as better drainage systems to cope with increased rainfall and potential flooding. Mumbai is India’s most densely populated city with poor sewage and drainage systems. Left unmanaged, this will result in widespread flooding and the spread of diseases. This will impact Mumbai’s poorest residents the most, who often live in shanty towns like Dharavi. However, official mitigation plans do not include any outreach work in these unplanned settlements to educate residents on how to tackle the coming changes.

India has undertaken massive tree-planting campaigns, like the Green India Mission, to combat deforestation and absorb atmospheric carbon dioxide. These specific actions showcase India’s commitment to adapting to and mitigating the impacts of climate change in a sustainable way. Nevertheless, the challenge for India will be balancing its desire to develop economically whilst also meeting its emissions targets.

Table 5.2: A timeline of India’s actions to sustainably manage the impacts of climate change

2008	India launches the National Action Plan on Climate Change (NAPCC), outlining a strategy to address climate change through eight national missions.
2014	Under the NAPCC, the Green India Mission focuses on increasing forest and tree cover and promoting sustainable forestry practices.
2015	India plays a crucial role in the negotiations leading to the Paris Agreement, committing to reduce its carbon intensity and increase the share of its non-fossil energy capacity.
2016	India co-founds the International Solar Alliance to promote solar energy and co-operation among solar-rich countries.
2019	India announces the National Mission on Transformative Mobility and Battery Storage, which plans to promote electric vehicles, introducing incentives for manufacturers and consumers of electric vehicles.
2020	Ambitious targets are set for expanding India’s renewable energy capacity, with a focus on solar and wind power.
2021	India participates in the COP26 summit, showcasing its efforts and commitment to addressing climate change on the global stage.
2022	India continues implementing various measures outlined in the NAPCC, including renewable energy projects, afforestation programs, and initiatives to enhance energy efficiency.
2023	At COP28, India commits to a reduction in emissions intensity, increasing non-fossil fuel electricity generation capacity, and the creation of new carbon sinks such as forests.

TASK 3

a

What do you think will be the hardest climate-related challenge for India to overcome? Share your ideas with the class.

b

Discuss in pairs who will benefit from the introduction of electric vehicles in India.

c

‘India is not on track to deal with the impacts of climate change’. With the aid of examples, how far do you agree with this statement?



Figure 5.17: Tree planting is a sustainable approach helping India to reduce the impacts of climate change

IMPROVE THIS ANSWER

For a named example you have studied, describe the effects of climate change.

India 1 accounts for around 7% of the world's global greenhouse gas emissions and is feeling the effects of climate change already.

The country has experienced a rise in average temperatures, 2 leading to more frequent and intense heatwaves. This has impacted agriculture, water resources, and public health. 3 In 2022 a heatwave caused temperatures to reach over 40°C in India. 4

The Himalayan region, which is a crucial source of freshwater for many rivers in India, has witnessed glacial retreat in recent years. At least one-third of the glaciers in the region will be gone by 2100. This can affect agriculture and the water supplies of urban areas downstream. 5

Changing monsoon patterns are resulting in variations in rainfall distribution and intensity. This can affect agriculture, as farmers rely heavily on the monsoon for water. India is projected to experience a 20% increase in extreme rainfall events which can wash away topsoil and nutrients due to erosion. 6

1 A named example is required, and the answer has introduced this already. This introduction is brief but gives some relevant background information, which shows that the topic is understood.

2 Increased temperatures is the first effect mentioned.

3 The answer suggests what some of the effects of rising temperatures will be.

4 Place-specific detail is included here using the example.

5 A second paragraph with a slightly different structure. Making a point, giving place-specific detail, and then developing the answer.

6 A third paragraph in a similar style shows that this answer has strong subject knowledge. This response has three well-developed points, with place-specific detail, so this is an excellent answer to the question. Note that the answer does not give reasons for the impacts, as that would be explaining, which is not what the question is asking.

Now, write your own response to this answer using the guidance above.

SELF-EVALUATION CHECKLIST

After studying this topic, think about how confident you are with the different topics. This will help you to find any gaps in your knowledge and learn more effectively.

I can:	Needs more work	Getting there	Ready to move on
outline the natural and human causes of climate change			
explain the effects of climate change at a local, regional, and global scale			
explain the reasons for and the impacts of the globalisation of food and energy supplies.			



› Theme 1

Practice questions

> Topic 1 Rivers

1

- a Study Figure 1.1, a map of the drainage basin of the Mississippi River in the USA.

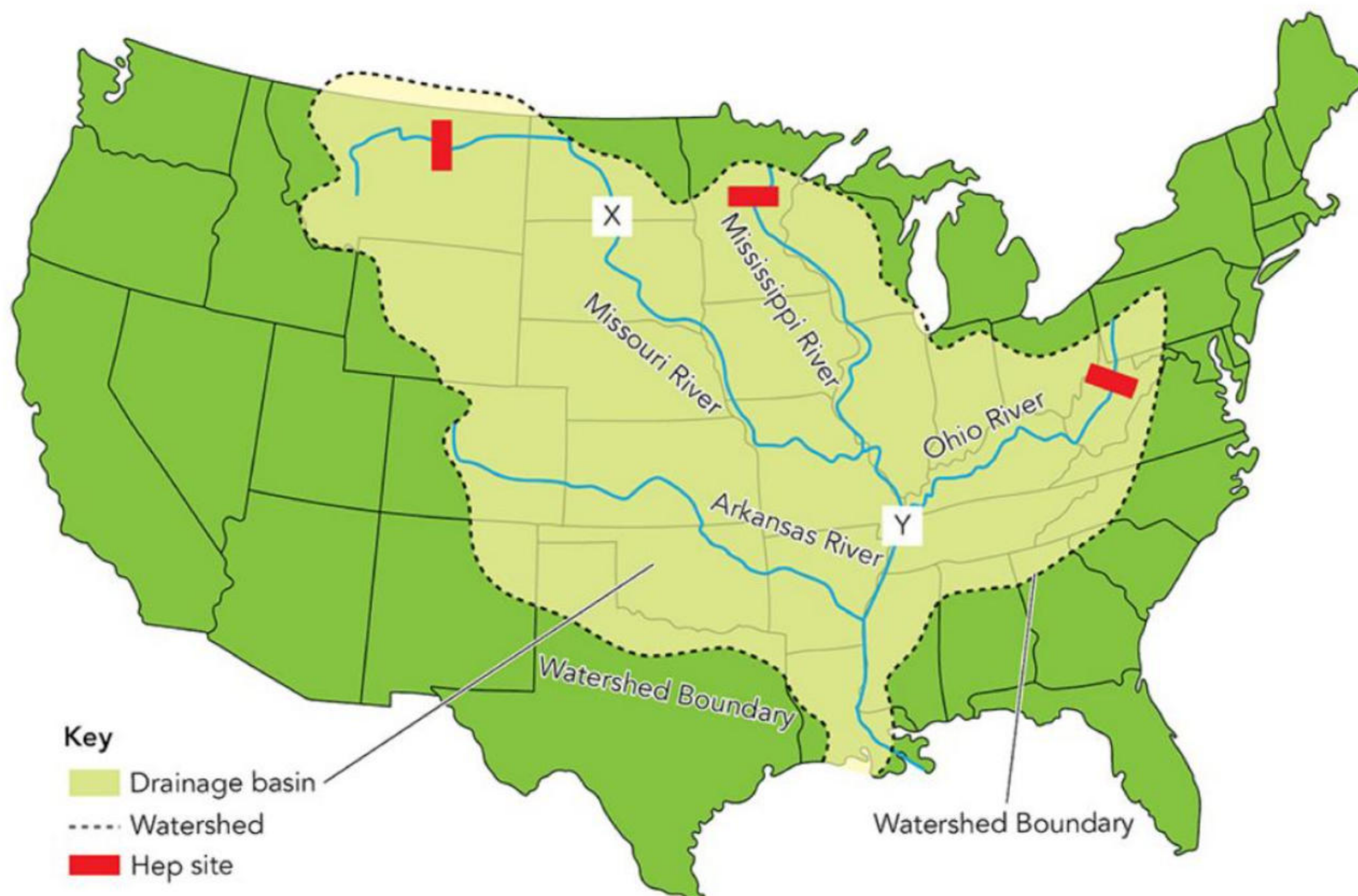


Figure 1.1

- i In what general direction does the river flow between X and Y? [1]
- ii Mark the following on Figure 1.1:
 - iii an **M** on the mouth of the Mississippi River
 - iv a **C** on a confluence. [2]
- v Suggest **three** likely differences between the river valleys at X and Y. [3]
- vi There are many hydroelectric power projects along the Mississippi River. Suggest reasons why hydroelectric power can be generated at the sites shown in Figure 1.1. [4]

- b Study Figure 1.2, a photograph of meanders and oxbow lakes.



Figure 1.2

-
- i Describe the features of the oxbow lake shown in Figure 1.2. [3]
- ii Suggest how the oxbow lake shown in Figure 1.2 was formed. [5]
- c Describe the main features of a waterfall and explain its formation. You may use a labelled diagram or diagrams. [7]

> Topic 2 Coastal environments

2

- a** Study Figure 2.1, which shows a coastline in England.

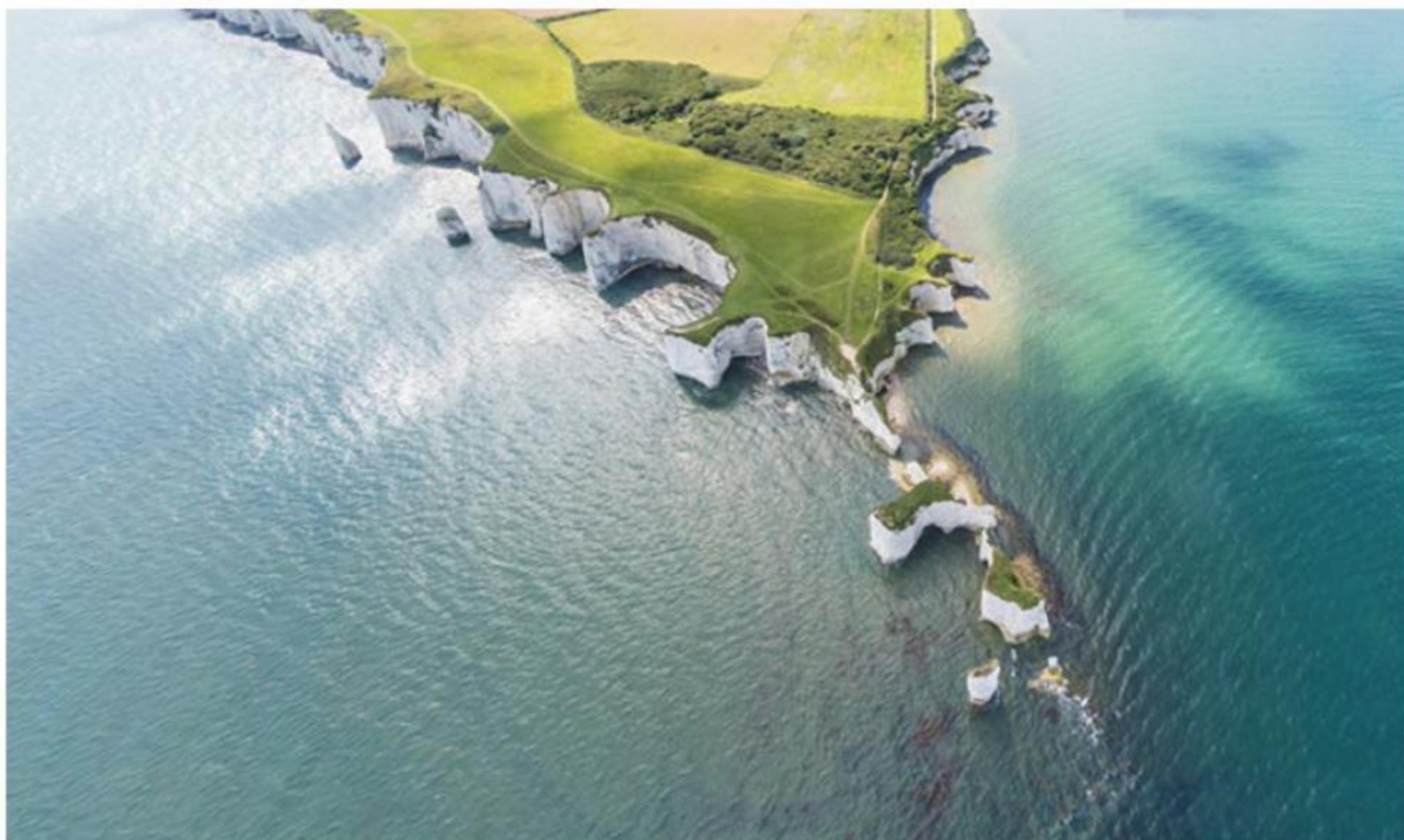


Figure 2.1

- | | | |
|------------|---|------------|
| i | Define <i>hydraulic action</i> . | [1] |
| ii | Identify two coastal landforms shown in Figure 2.1. | [2] |
| iii | Suggest how one of the coastal landforms you identified in Figure 2.1 was formed. | [3] |
| b | Explain two methods that could be used to reduce the impact of coastal erosion along a stretch of coastline. | [4] |
| c | Describe three threats to coral reefs. | [3] |
| d | Draw an annotated diagram that shows the process of longshore drift. | [5] |
| e | For a named area you have studied, explain how coastal erosion is being managed. | [7] |

> Topic 3 Changing ecosystems

3

a Study Figure 3.1, showing the distribution of Weddell seals in Antarctica.

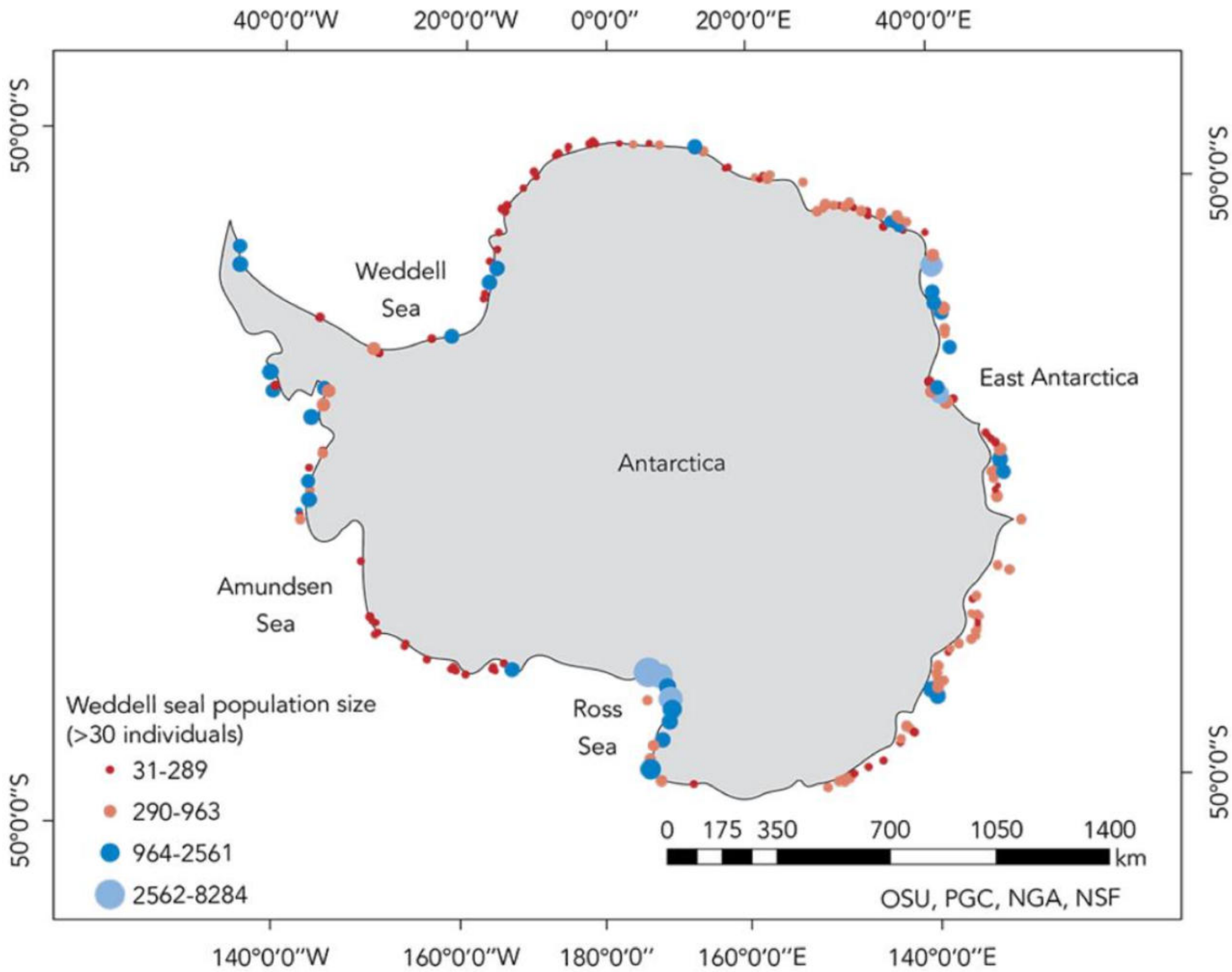


Figure 3.1

- i

Describe the distribution of Weddell seals in Antarctica.

[2]
- ii

Suggest two reasons for the distribution of Weddell seals.

[2]
- iii

Both biotic and abiotic factors influence the distribution of organisms. What is meant by the term *abiotic*?

[1]
- iv

Describe the climate of Antarctica.

[3]
- v

Table 3.1 shows the estimated number of Weddell seals in different regions in Antarctica.

Table 3.1

Region	Estimated number of Weddell seals	Total % by region
1	4,926	5
2	10,405	12
3	10,975	11
4	14,038	
5	31,415	
6	13,586	16
Total	85,345	100

Calculate the percentage of Weddell seals found in Region 4 and 5.

[2]

vi Overfishing is a threat to resources in Antarctica. Suggest how overfishing could impact the Weddell seal population.

[2]

b Study Figure 3.2, which shows climate graphs for two places.

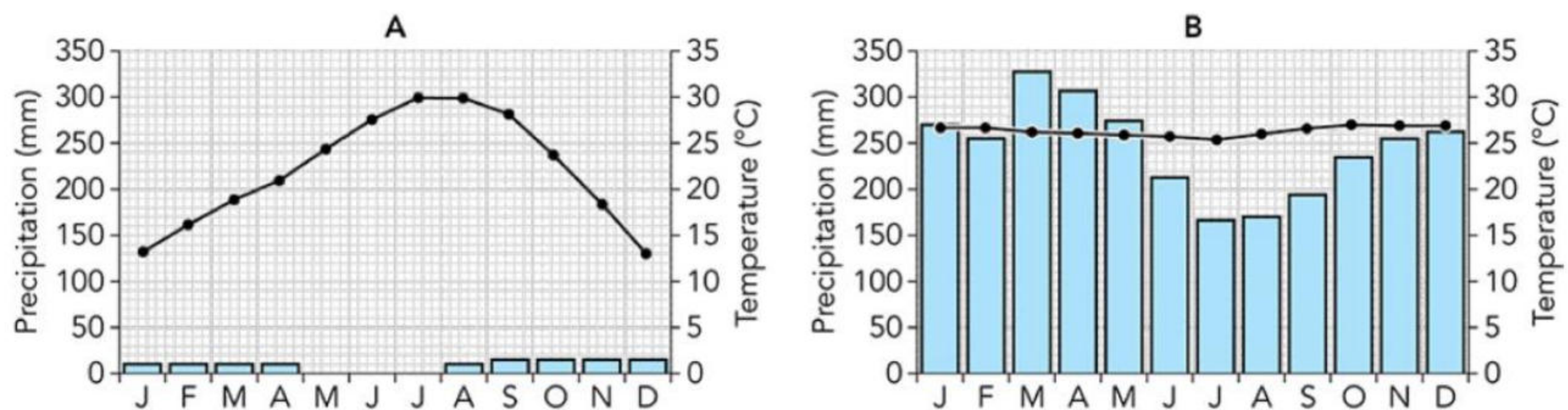


Figure 3.2

i Identify the graph which shows a tropical rainforest climate. Give **two** reasons for your choice.

[3]

ii Explain why some areas have a tropical rainforest climate.

[3]

c For a named ecosystem you have studied, describe strategies that can be used to manage the ecosystem sustainably.

[7]

> Topic 4 Tectonic hazards

4

a Study Figure 4.1, showing two different types of plate boundaries.

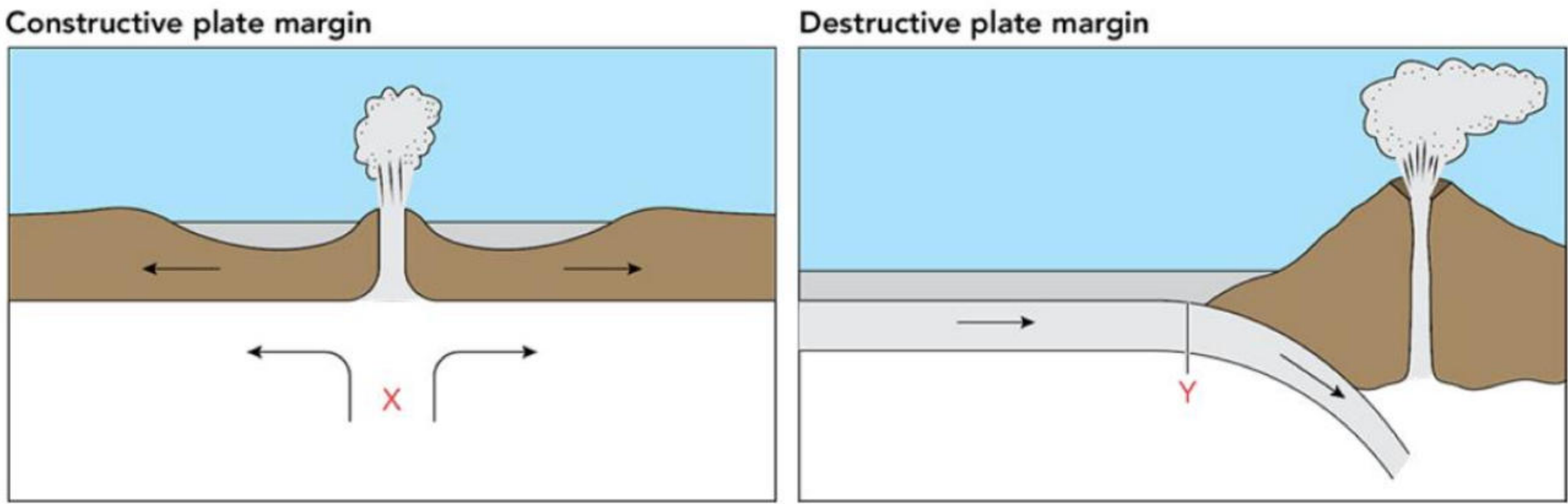


Figure 4.1

► Image Description

- i Name the type of volcano most likely to form on a destructive plate boundary. [1]
- ii Compare the direction of movement of plates at a constructive and a destructive plate boundary. [1]
- iii Choose the correct label to be placed at X and Y in Figure 4.1. [2]

Convection currents Focus Oceanic trench

Ash cloud Continental plate

X

Y
- iv Explain why the type of volcano you identified in Question 4ai forms on a destructive plate boundary. [4]

b Study Figure 4.2, showing the frequency of magnitude 5+ earthquakes worldwide between 2000 and 2023

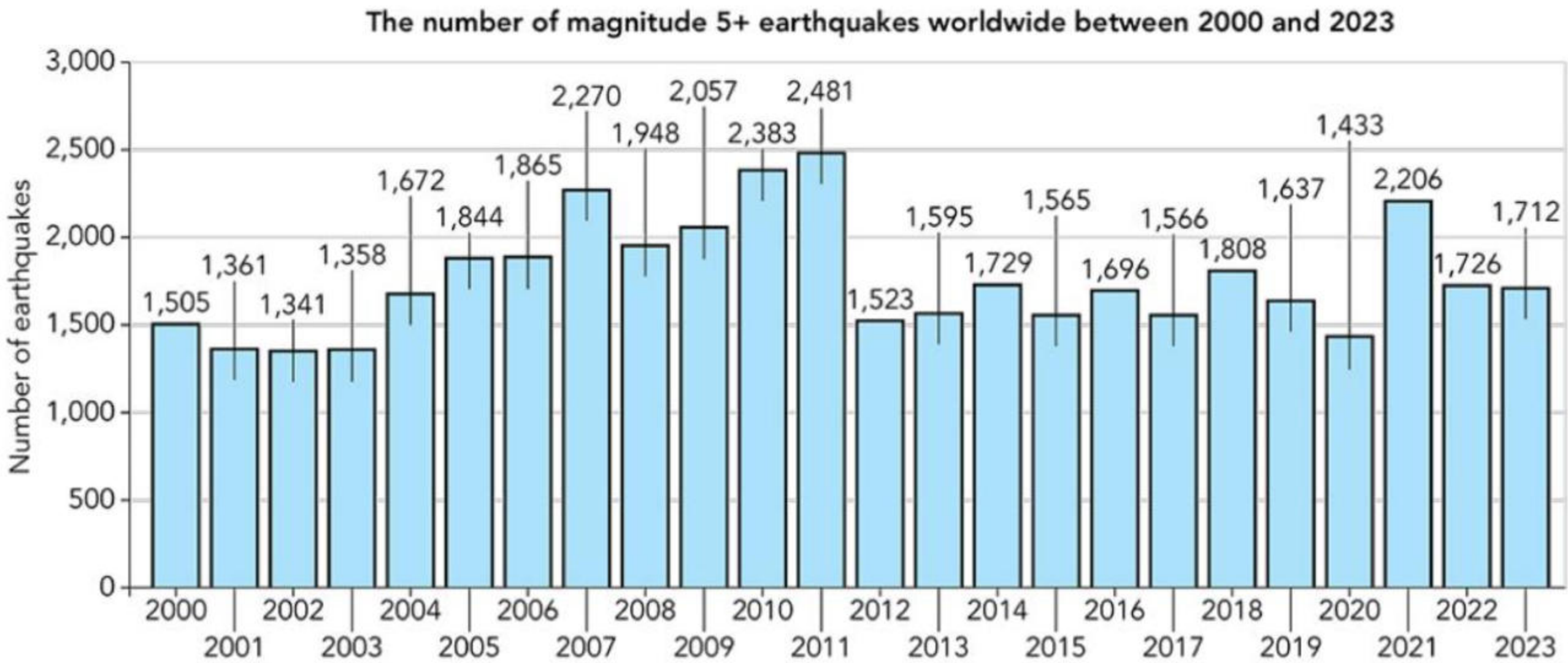


Figure 4.2

- i Which year recorded the highest number of earthquakes (M5+)? [1]
- ii Describe the changes in the number of earthquakes (M5+) between 2000 and 2023. You should use data to support your answer. [3]
- iii In 1950, 392 earthquakes of M5+ were recorded. Explain why so few earthquakes were recorded in 1950 compared to 2023. [2]
- iv Suggest reasons why earthquakes in a Low Income Country have greater impacts than earthquakes in a High Income Country. [4]

c For a named volcanic eruption that you have studied, explain the effects of the eruption.

[7]

> Topic 5 Climate change

5

- a Study Table 5.1, which shows global average surface temperatures between 2010 and 2023.

Table 5.1

Year	Differences from 1901–2000 average (°C)
2010	0.73
2011	0.63
2012	0.66
2013	0.68
2014	0.77
2015	0.92
2016	1.03
2017	0.95
2018	0.86
2019	0.98
2020	1.01
2021	0.86
2022	0.91
2023	1.18

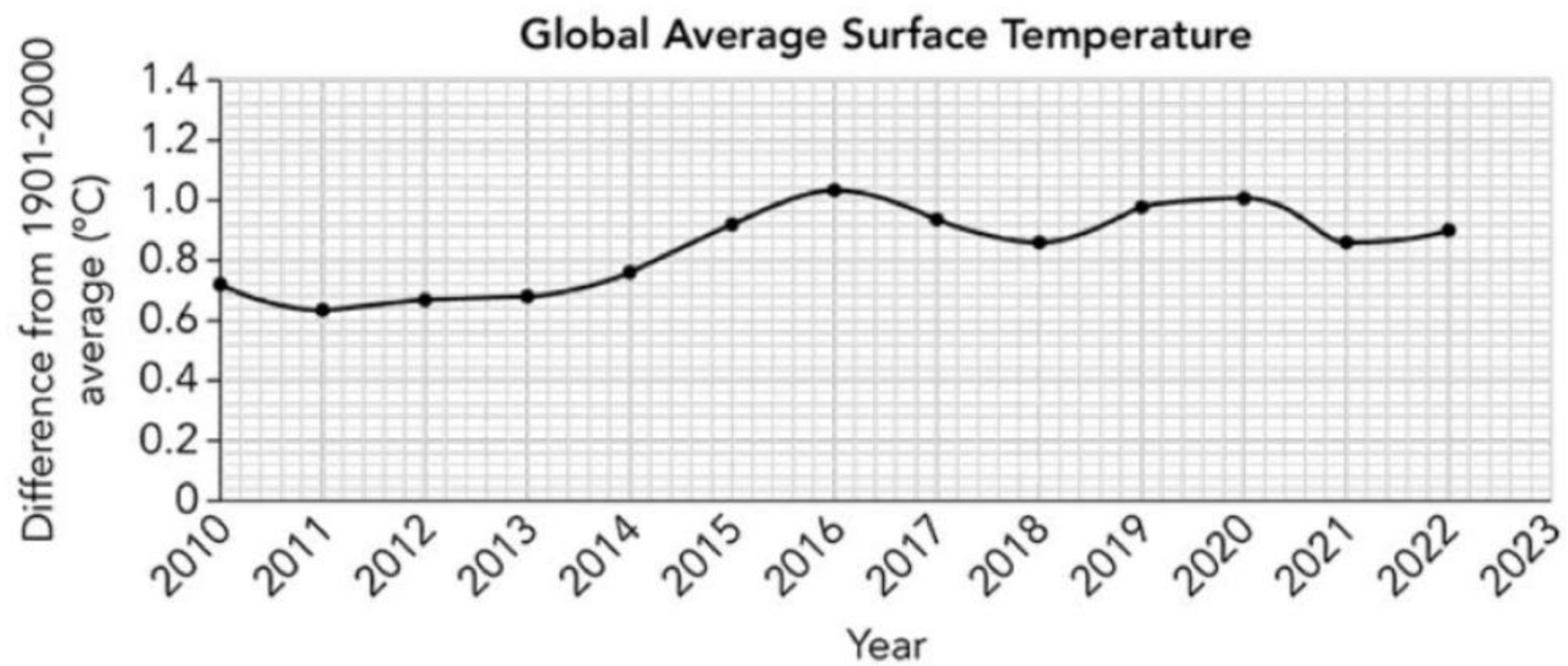


Figure 5.1

► Image Description

- i Define *global warming*. [1]
 - ii Complete the graph in Figure 5.1 using the data in Table 5.1. [2]
 - iii Describe the graph shown in Figure 5.1. [3]
 - iv Suggest reasons for the patterns shown in Figure 5.1. [4]
- b Study Figure 5.2, which shows a melting ice sheet in Antarctica.



Figure 5.2

-
- | | | |
|-----------|---|------------|
| i | Outline three impacts of melting ice sheets and glaciers. | [3] |
| ii | Explain what people can do to reduce the impacts of climate change. | [5] |
| c | For a named country you have studied, explain what has been done to reduce the impacts of climate change. | [7] |

› Theme 2

Human Geography

> Topic 6

Population change

LEARNING INTENTIONS

By the end of this topic you will be able to:

- describe and give reasons for the rapid increase in the world's population
- understand the main causes of growth and decline in a country's population
- understand and evaluate the demographic transition model
- identify and evaluate the implications of different types of population structures
- explain and give reasons for population migration
- understand some of the main impacts of migration and how migration is managed.

GETTING STARTED

Working in pairs, take 10 minutes to discuss the following questions:

- 1 Many governments carry out a census, or count, every few years to get an idea of the structure of their population. Why do you think this is a valuable survey for the government of a country?
- 2 International migration is a 'hot' geographical topic. Brainstorm the advantages and disadvantages of migration for both the country of origin and the destination country.